

Let

$$N = 2025^2$$

be the total number of cells.

0.0.1 a) The game always terminates

The number of grey cells never increases. Moreover, on every one of Sue's turns, her chosen 2×2 block contains at least one grey cell, and all four cells are painted scarlet. Hence the number of grey cells decreases by at least 1 on every Sue turn.

Consequently, Sue can make at most N moves. Since the players alternate, the entire game has at most $2N$ moves and therefore always ends after finitely many moves.

0.0.2 b) The optimal number of lime cells

Set

$$m = \frac{2025 + 1}{2} = 1013.$$

We will prove that the value of the game is

$$m^2 - 1 = 1013^2 - 1 = 1,026,168.$$

Linda can guarantee at least $m^2 - 1$ lime cells Let

$$P = (r, c) : r \text{ and } c \text{ are both odd.}$$

There are 1013 odd-numbered rows and 1013 odd-numbered columns, so

$$|P| = 1013^2 = m^2.$$

Every axis-aligned 2×2 block contains exactly one cell of P , since two consecutive rows contain exactly one odd row and two consecutive columns contain exactly one odd column.

Linda uses the following strategy:

1. On her first turn, she paints any cell of P lime.
2. After each nonterminal move by Sue, let p be the unique cell of P in the 2×2 block just painted by Sue. Linda repaints p lime.

After Linda's first turn, all her moves repaint a cell that Sue has just painted scarlet, so these moves do not remove any grey cells. Thus the eventual terminal move is made by Sue.

Consider the final 2×2 block painted by Sue, and let p_{last} be its unique cell in P . This cell is scarlet at the end. We claim that every other cell of P is lime.

Indeed, take $p \in P \setminus p_{\text{last}}$.

- If Sue never painted p , then p must be the cell Linda painted on her first turn; otherwise it would still be grey when the game ended. Hence it is lime.
- If Sue painted p , consider the last Sue move whose block contained p . This was not the terminal move, because $p \neq p_{\text{last}}$. Linda therefore immediately repainted p lime, and no later Sue move affected it.

Thus all $m^2 - 1$ cells of $P \setminus p_{\text{last}}$ are lime. Therefore Linda can guarantee at least

$$m^2 - 1$$

lime cells.

Sue can guarantee at most $m^2 - 1$ lime cells Construct the following family of row-pairs:

$$1, 2, 3, 4, \dots, 2023, 2024, 2024, 2025.$$

There are $1013 = m$ such pairs, and together they cover every row. Construct the analogous m column-pairs.

Taking every Cartesian product of one row-pair and one column-pair gives a family $\mathcal{B}$ of

$$m^2$$

axis-aligned 2×2 blocks whose union covers the entire board.

Sue plays as follows.

- After Linda's first move, Sue chooses a block of $\mathcal{B}$ containing Linda's chosen cell and paints it scarlet. This is legal because the other three cells of the block are still grey.
- Thereafter, whenever grey cells remain, Sue chooses any grey cell and paints an unused block of $\mathcal{B}$ containing it.

Such an unused block always exists. Indeed, every cell belongs to at least one block of $\mathcal{B}$. A grey cell cannot belong to a block that Sue has already used, because when that block was used the cell became scarlet, and cells never become grey again.

Thus Sue uses each block of $\mathcal{B}$ at most once, so she makes at most m^2 moves. Furthermore, Linda's very first lime mark is immediately overwritten.

Let T be the total number of Linda moves. Every cell that is lime at the end has a distinct last Linda move on which it was painted. None of these last moves can be Linda's first move, since that mark was overwritten immediately. Hence

$$\# \text{final lime cells} \leq T - 1.$$

There are two possible ways the game ends.

- If it ends after Sue's s -th move, then Linda has made $T = s$ moves and $s \leq m^2$. Hence

$$\# \text{lime cells} \leq s - 1 \leq m^2 - 1.$$

- If it ends after Linda's move following Sue's s -th move, then $T = s + 1$. In this case $s \leq m^2 - 1$, because if Sue had already used all m^2 blocks, their union would cover the board and the game would have ended on Sue's preceding move. Therefore

$$\# \text{lime cells} \leq T - 1 = s \leq m^2 - 1.$$

So Sue can ensure that there are at most $m^2 - 1$ lime cells.

The two bounds coincide. Therefore, under optimal play, the final number of lime cells is

$$\boxed{1013^2 - 1 = \boxed{1,026,168}}.$$